

Session Info

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Session

```
sessionInfo()  
  
#> R version 4.5.2 (2025-10-31)  
#> Platform: x86_64-pc-linux-gnu  
#> Running under: Ubuntu 24.04.4 LTS  
#>  
#> Matrix products: default  
#> BLAS: /usr/lib/x86_64-linux-gnu/openblas-pthread/libblas.so.3  
#> LAPACK: /usr/lib/x86_64-linux-gnu/openblas-pthread/libopenblasp-r0.3.26.so; LAPACK version 3.12.0  
#>  
#> locale:  
#>  [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C  
#>  [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8  
#>  [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8  
#>  [7] LC_PAPER=en_US.UTF-8     LC_NAME=C  
#>  [9] LC_ADDRESS=C             LC_TELEPHONE=C  
#> [11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C  
#>  
#> time zone: Etc/UTC
```

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#> tzcode source: system (glibc)
#>
#> attached base packages:
#> [1] stats      graphics  grDevices  utils      datasets  methods    base
#>
#> other attached packages:
#> [1] rProject_0.0.25
#>
#> loaded via a namespace (and not attached):
#> [1] backports_1.5.0  R6_2.6.1          lubridate_1.9.5  xfun_0.56
#> [5] magrittr_2.0.4   glue_1.8.0         stringr_1.6.0    knitr_1.51
#> [9] timechange_0.4.0 generics_0.1.4     lifecycle_1.0.5  xml2_1.5.2
#> [13] cli_3.6.5.9000   bibtex_0.5.2       compiler_4.5.2    plyr_1.8.9
#> [17] highr_0.12       rprojroot_2.1.1    httr_1.4.8        tools_4.5.2
#> [21] evaluate_1.0.5   Rcpp_1.1.1         RefManageR_1.4.0  otel_0.2.0
#> [25] rlang_1.1.7      jsonlite_2.0.0     stringi_1.8.7

```

Packages

```

unname(installed.packages()[, 1])

#> [1] "fitVARMxID"      "metaDyn"          "OpenMx"
#> [4] "simStateSpace"   "abind"             "Amelia"
#> [7] "arm"             "arrow"             "ash"
#> [10] "AsioHeaders"     "askpass"           "assertthat"
#> [13] "backports"       "base64enc"         "betaDelta"
#> [16] "betaMC"          "betaNB"            "betaSandwich"

```

```

#> [19] "BH" "bibtex" "BiocManager"
#> [22] "BiocVersion" "bit" "bit64"
#> [25] "bitops" "blob" "bootStateSpace"
#> [28] "brew" "brio" "broom"
#> [31] "bslib" "cachem" "callr"
#> [34] "car" "caracas" "carData"
#> [37] "cellranger" "cfr" "checkmate"
#> [40] "cli" "clipr" "clock"
#> [43] "clusterGeneration" "coda" "c0de"
#> [46] "collections" "colorspace" "commonmark"
#> [49] "conflicted" "corpcor" "covr"
#> [52] "cowplot" "cpp11" "crayon"
#> [55] "credentials" "crosstalk" "cTMed"
#> [58] "ctsem" "curl" "data.table"
#> [61] "DBI" "dbplyr" "Deriv"
#> [64] "desc" "deSolve" "devtools"
#> [67] "diagram" "dials" "DiceDesign"
#> [70] "diffobj" "digest" "distributional"
#> [73] "distro" "doBy" "docopt"
#> [76] "downlit" "dplyr" "DT"
#> [79] "dtplyr" "duckdb" "dynr"
#> [82] "dynUtils" "ellipse" "ellipsis"
#> [85] "evaluate" "expm" "fansI"
#> [88] "farver" "fastDummies" "fastmap"
#> [91] "fclust" "fda" "fdrtool"
#> [94] "fds" "fitVARMxID" "FNN"
#> [97] "fontawesome" "forcats" "foreach"

```

#> [100]	"forecast"	"Formula"	"fracdiff"
#> [103]	"fs"	"fst"	"fstcore"
#> [106]	"furrr"	"future"	"future.apply"
#> [109]	"gargle"	"generics"	"gert"
#> [112]	"ggplot2"	"ggrepel"	"gh"
#> [115]	"gitcreds"	"glasso"	"glmnet"
#> [118]	"globals"	"glue"	"googledrive"
#> [121]	"googlesheets4"	"gower"	"GPARotation"
#> [124]	"GPfit"	"graphicalVAR"	"gridExtra"
#> [127]	"gsubfn"	"gtable"	"gtools"
#> [130]	"hardhat"	"haven"	"hdcrcde"
#> [133]	"here"	"highr"	"Hmisc"
#> [136]	"hms"	"htmlTable"	"htmltools"
#> [139]	"htmlwidgets"	"httpgd"	"httpuv"
#> [142]	"httr"	"httr2"	"ids"
#> [145]	"ifaTools"	"igraph"	"infer"
#> [148]	"ini"	"inline"	"ipred"
#> [151]	"isoband"	"iterators"	"jomo"
#> [154]	"jpeg"	"jquerylib"	"jsonlite"
#> [157]	"jsonvalidate"	"kernlab"	"knitr"
#> [160]	"ks"	"kutils"	"labeling"
#> [163]	"Lahman"	"languageserver"	"later"
#> [166]	"latex2exp"	"lava"	"lavaan"
#> [169]	"lazyeval"	"lhs"	"lifecycle"
#> [172]	"lintr"	"lisrelToR"	"listenv"
#> [175]	"litedown"	"littler"	"lme4"
#> [178]	"lmtest"	"locfit"	"longMI"

#> [181]	"loo"	"lubridate"	"magick"
#> [184]	"magrittr"	"markdown"	"MatrixModels"
#> [187]	"matrixStats"	"mclust"	"memoise"
#> [190]	"metaDyn"	"metaSEM"	"mi"
#> [193]	"mice"	"microbenchmark"	"mime"
#> [196]	"miniUI"	"minqa"	"mitml"
#> [199]	"mize"	"mlVAR"	"mnormt"
#> [202]	"modeldata"	"modelenv"	"modelr"
#> [205]	"MplusAutomation"	"multicool"	"mvtnorm"
#> [208]	"nloptr"	"numDeriv"	"nycflights13"
#> [211]	"OpenMx"	"openssl"	"openxlsx"
#> [214]	"ordinal"	"otel"	"pak"
#> [217]	"pan"	"pander"	"parallelly"
#> [220]	"parsnip"	"patchwork"	"pbapply"
#> [223]	"pbivnorm"	"pbkrtest"	"pcaPP"
#> [226]	"pdftools"	"pillar"	"pkgbuild"
#> [229]	"pkgconfig"	"pkgdown"	"pkgload"
#> [232]	"plyr"	"png"	"posterior"
#> [235]	"pracma"	"praise"	"prettyunits"
#> [238]	"printr"	"processx"	"prodlim"
#> [241]	"profvis"	"progress"	"progressr"
#> [244]	"promises"	"proto"	"ps"
#> [247]	"psych"	"purrr"	"qgraph"
#> [250]	"qpdf"	"quadprog"	"quantreg"
#> [253]	"quarto"	"QuickJSR"	"R.cache"
#> [256]	"R.methodsS3"	"R.oo"	"R.utils"
#> [259]	"R2jags"	"R2WinBUGS"	"R6"

#> [262]	"ragg"	"rainbow"	"rappdirs"
#> [265]	"rbibutils"	"rcmdcheck"	"RColorBrewer"
#> [268]	"Rcpp"	"RcppArmadillo"	"RcppEigen"
#> [271]	"RcppGSL"	"RcppParallel"	"RcppTOML"
#> [274]	"RCurl"	"Rdpack"	"readr"
#> [277]	"readxl"	"recipes"	"RefManageR"
#> [280]	"reformulas"	"rematch"	"rematch2"
#> [283]	"remotes"	"reprex"	"reshape2"
#> [286]	"reticulate"	"rex"	"rhdf5"
#> [289]	"rhdf5filters"	"Rhdf5lib"	"rhub"
#> [292]	"rjags"	"rlang"	"RMariaDB"
#> [295]	"rmarkdown"	"rockchalk"	"roxygen2"
#> [298]	"rpf"	"RPostgres"	"rProject"
#> [301]	"rprojroot"	"rsample"	"RSQLite"
#> [304]	"rstan"	"rstantools"	"rstudioapi"
#> [307]	"RUnit"	"rversions"	"rvest"
#> [310]	"Ryacas"	"S7"	"sass"
#> [313]	"scales"	"selectr"	"sem"
#> [316]	"semlbci"	"semmcci"	"semPlot"
#> [319]	"sessioninfo"	"sfd"	"shape"
#> [322]	"shiny"	"simStateSpace"	"slider"
#> [325]	"snow"	"snowfall"	"sourcetools"
#> [328]	"SparseM"	"sparsevctrs"	"SQUAREM"
#> [331]	"StanHeaders"	"stringi"	"stringr"
#> [334]	"styler"	"symSEM"	"sys"
#> [337]	"systemfonts"	"tailor"	"tensorA"
#> [340]	"testthat"	"texreg"	"textshaping"

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#> [343] "tibble"          "tidymodels"      "tidyr"
#> [346] "tidyselect"      "tidyverse"       "timechange"
#> [349] "timeDate"        "tinytex"         "tune"
#> [352] "tzdb"            "ucminf"          "unigd"
#> [355] "urca"            "urlchecker"      "usethis"
#> [358] "utf8"            "uuid"            "V8"
#> [361] "vctrs"           "viridisLite"     "vroom"
#> [364] "waldo"           "warp"            "whisker"
#> [367] "whoami"          "withr"           "workflows"
#> [370] "workflowsets"    "xfun"            "XML"
#> [373] "xml2"            "xmlparsedata"    "xopen"
#> [376] "xtable"          "yaml"            "yardstick"
#> [379] "zip"             "zoo"             "base"
#> [382] "boot"            "class"           "cluster"
#> [385] "codetools"       "compiler"        "datasets"
#> [388] "foreign"         "graphics"        "grDevices"
#> [391] "grid"            "KernSmooth"      "lattice"
#> [394] "MASS"            "Matrix"          "methods"
#> [397] "mgcv"            "nlme"            "nnet"
#> [400] "parallel"        "rpart"           "spatial"
#> [403] "splines"         "stats"           "stats4"
#> [406] "survival"        "tcltk"           "tools"
#> [409] "utils"

```

References

R Core Team. (2025). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. <https://www.R-project.org/>